

2024-25 秋春概率统计标准答案和评分标准

一、 1)B, 2)B, 3)B, 4)A, 5)C

二、

1) $2/7=0.286$; 2) $e^{-4} = 0.0183$; 3) $1/3$; 4) 0.0228 ; 5) 0 ; 6) -3 ; 7) 5 ; 8) $1/5$

$$9) F(x) = \begin{cases} 0 & x < 1 \\ 0.3 & 1 \leq x < 2 \\ 0.8 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$10) f_Y(y) = \begin{cases} 1.5\sqrt{y} & 0 < y < 1 \\ 0 & \text{其它} \end{cases}$$

11) $1/2$; 12) $[0.9, 3.1]$; 13) 5

三、 1)

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \dots\dots\dots 1'$$

$$\int_0^1 \int_0^y ax dx dy = 1 \dots\dots\dots 2'$$

$$a = 6 \dots\dots\dots 2'$$

$$2) f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \dots\dots\dots 1'$$

$$\text{当 } x < 0, \text{ or } x > 1 \text{ 时, } f_X(x) = 0; \dots\dots\dots 1'$$

$$\text{当 } 0 < x < 1 \text{ 时,}$$

$$f_X(x) = \int_x^1 6xy = 6x(1-x) \dots\dots\dots 3'$$

$$3) f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} \dots\dots\dots 1'$$

$$= \begin{cases} \frac{1}{1-x} & x < y < 1 \\ 0 & \text{其他} \end{cases}, (0 < x < 1) \dots\dots\dots 1'$$

$$f_{Y|X}(y|0.5) = \begin{cases} 2 & 0.5 < y < 1 \\ 0 & \text{其他} \end{cases}, \dots\dots\dots 1$$

$$P(0.3 < Y < 0.6 | X = 0.5) = \int_{0.5}^{0.6} 2 dy \dots\dots\dots 1'$$

$$= 0.2 \dots\dots\dots 1'$$



四、

B - 第二次取到的球为白球

A_i - 第一次取出两球中包含 i 个红球; $i = 0, 1, 2$;

$$P(A_0) = \frac{C_2^2}{C_5^2} = \frac{1}{10}; P(A_1) = \frac{C_2^1 C_3^1}{C_5^2} = \frac{6}{10};$$

$$P(A_2) = \frac{C_3^2}{C_5^2} = \frac{3}{10}, \dots\dots\dots 1'$$

$$P(B|A_0) = 0; P(B|A_1) = \frac{1}{5}; P(B|A_2) = \frac{2}{5}; \dots\dots\dots 1'$$

$$(1) P(B) = P(A_0)P(B|A_0) + P(A_1)P(B|A_1) + P(A_2)P(B|A_2) \dots\dots\dots 2'$$

$$= 0.24 = 6/25 \dots\dots\dots 2'$$

$$(2) P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(B)} \dots\dots\dots 2'$$

$$= \frac{\frac{6}{10} \times \frac{1}{5}}{\frac{6}{25}} = 0.5 \dots\dots\dots 2'$$

五 方法一

$$f_X(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}; f_Y(y) = \begin{cases} 2e^{-2y} & y \geq 0 \\ 0 & y < 0 \end{cases} \dots\dots\dots 1'$$

$$f(x, y) = f_X(x)f_Y(y) \dots\dots\dots 2'$$

$$F(z) = P(Z \leq z) = P(X + Y \leq z) \dots\dots\dots 1'$$

$$\text{当 } z < 0 \text{ 时, } F(z) = 0; \dots\dots\dots 1'$$

$$\text{当 } z \geq 0 \text{ 时, } F(z) = \int_0^z \int_0^{z-y} 2e^{-x-2y} dx dy = 1 - 2e^{-z} + e^{-2z} \dots\dots\dots 3'$$

$$f(z) = [F(z)]' \dots\dots\dots 1'$$

$$= \begin{cases} 2e^{-z} - 2e^{-2z} & z \geq 0 \\ 0 & \text{其他} \end{cases} \dots\dots\dots 1'$$

方法二

$$f_X(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}; f_Y(y) = \begin{cases} 2e^{-2y} & y \geq 0 \\ 0 & y < 0 \end{cases} \dots\dots\dots 1'$$

$$f(x, y) = f_X(x)f_Y(y) \dots\dots\dots 2'$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)dx \dots\dots\dots 3'$$

$$\text{当 } z < 0 \text{ 时, } f_Z(z) = 0; \dots\dots\dots 1'$$

$$\text{当 } z \geq 0 \text{ 时, } f_Z(z) = \int_0^z e^{-x} 2e^{-2(z-x)} dx \dots\dots\dots 2'$$

$$= 2e^{-z} - 2e^{-2z} \dots\dots\dots 1'$$



六、

记 $\mu = EX = 0; \sigma^2 = DX = \frac{1}{3}, \dots \dots \dots 2'$

记 $X_1, \dots, X_n$ 为来自该总体的样本, $n = 300$

所求概率为

$$p = P(\sum_{i=1}^n X_i \leq 5) \dots \dots \dots 2'$$

$$\approx \Phi\left(\frac{5-n\mu}{\sqrt{n}\sigma}\right) = \Phi\left(\frac{1}{2}\right) \dots \dots \dots 5'$$

$$\mu = EX = np = 90; \sigma^2 = \text{var}(X) = np(1-p) = 9; \dots \dots \dots 2'$$

七、

$$(1) \text{似然函数为: } L(\theta) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \frac{1}{2} e^{-(X_i-\theta)/2} \dots \dots \dots 1'$$

$$= 2^{-n} e^{-\frac{X_1+X_2+\dots+X_n}{2}} e^{n\theta/2}, \theta \leq \min(X_1, \dots, X_n) \dots \dots \dots 2'$$

$L(\theta)$ 是 θ 的单调增函数, 所以, 当 $\theta = \min(X_1, \dots, X_n)$ 时, $L(\theta)$ 取最大值。 $\dots \dots \dots 2$

故 θ 的最大似然估计量为:

$$\hat{\theta} = \min(X_1, \dots, X_n) \dots \dots \dots 1'$$

$$(2) \text{总体的分布函数为: } F(x) = \begin{cases} 1 - e^{-\frac{x-\theta}{2}} & x \geq \theta \\ 0 & x < \theta \end{cases} \dots \dots \dots 1'$$

$\hat{\theta}$ 的概密度函数为, $F_{\hat{\theta}}(t) = n[1 - F(t)]^{n-1} f(t)$

$$= \begin{cases} \frac{n}{2} e^{-\frac{n(t-\theta)}{2}} & t \geq \theta \\ 0 & t < \theta \end{cases} \dots \dots \dots 1'$$

$$E\hat{\theta} = \int_{\theta}^{+\infty} t \frac{n}{2} e^{-\frac{n(t-\theta)}{2}} dt = \theta + \frac{2}{n} \dots \dots \dots 1'$$

$E\hat{\theta} \neq \theta, \hat{\theta}$ 不是 θ 的无偏估计 $\dots \dots \dots 1'$

八、

$$(1) \text{检验统计量: } T = \frac{\bar{X} - \mu_0}{S_n} \sqrt{n} \sim t(n-1) (H_0 \text{ 成立时})$$

$$\text{拒绝域: } D = \left\{ |T| > t_{\frac{\alpha}{2}}(n-1) \right\} \dots \dots \dots 3'$$

其中, $n = 25, \mu_0 = 45, S_n = 10, \alpha = 0.05, t_{\frac{\alpha}{2}}(n-1) = 2.064;$

$$T \text{ 的观测值: } T = \frac{44 - 45}{10} \sqrt{25} = -0.5 \dots \dots \dots 1'$$

$$|T| = 0.5 < 2.064 \dots \dots \dots 1'$$

所以, 不能拒绝原假设。 $\dots \dots \dots 1'$

$$2) \sigma^2 \text{ 的置信区间: } \left[\frac{(n-1)S_n^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}, \frac{(n-1)S_n^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)} \right] \dots \dots \dots 2'$$

$$\alpha = 0.05; S_n^2 = 100, \chi_{\frac{\alpha}{2}}^2(n-1) = 39.36, \chi_{1-\frac{\alpha}{2}}^2(n-1) = 12.4; \dots \dots \dots 1'$$

所求置信区间为: $[60.98, 193.55] \dots \dots \dots 1'$

